MUST Smart Exam

Dashboard design data and implementation guide

Misr University for Science and Technology - Graduation project. AI-assisted exam monitoring with cameras and a robot, with two roles: **Doctor** (supervises the exam) and **Administrator** (manages colleges, students, exams and rooms).

1. Summary

- The dashboard follows the MUST website identity: navy header and sidebar, green accents, the official seal, bold condensed uppercase headings, and a green line under the top bar.
- It is a finished template (**must_dashboard.html** + **login.html**) connected to your Flask app through a new database module (**admin_db.py**) and an updated **dashboard.py**.
- Alerts are always shown as *suspected - needs human review*. The system never accuses or penalises a student automatically.
- Your old pages **index.html** (operator console) and **monitor.html** were re-themed with the same colors and logo.

2. Colors

Measured values were sampled from your must.edu.eg screenshot and the logo file. Other values were chosen by me to support the brand and are easy to change in the CSS variables at the top of each template.

	Use	Value	Source
	Main navy: top bar, sidebar, login panel, buttons	#202f6a	Measured (site header)
	Brand green: active menu, highlights, titles	#4da964	Measured (site nav/heading)
	Logo green (building)	#00a751	Measured (logo)
	Lighter navy: KPI numbers, chart bars	#2c3f8f	Chosen
	Darker green: links (readable on white)	#2f8a49	Chosen
	Page background	#f3f5f9	Chosen
	Cards	#ffffff	Chosen
	Borders	#dde3ee	Chosen
	Text	#14213d	Chosen
	Muted text	#64718a	Chosen

Status colors (same everywhere)

	Use	Value	Source
	Normal / online (text #16a34a)	#dcfce7	Chosen
	Warning / suspected (text #d97706)	#fef3c7	Chosen
	Danger / needs review (text #dc2626)	#fee2e2	Chosen

Dark mode

Follows the device setting, with a manual toggle in the top bar. Background #0b1322, cards #13203a, borders #25375a, text #e8edf7, muted #93a3c0, sidebar #101a3f. Buttons switch to green #4da964 with dark text.

3. Fonts

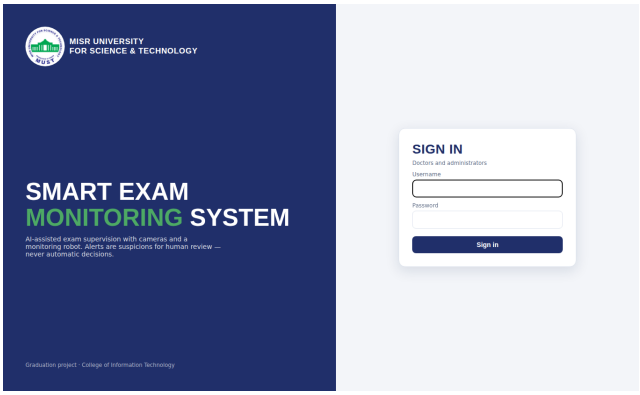
- **Headings** (page titles, menu, buttons, KPI numbers, card headings, table headers): Barlow Condensed 600/700 from Google Fonts, mostly uppercase. Fallbacks: Arial Narrow, Impact.
- **Body and tables**: Segoe UI, then system-ui, Roboto, Arial; 14 px.
- Barlow Condensed is my closest match to the site headline font. It is not confirmed - if you learn the real font name, change the --hf variable.

4. Logo

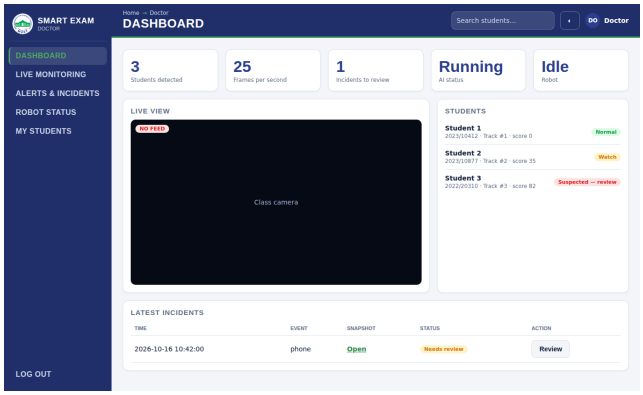
- The official MUST seal (your PNG, 778 px) resized to 160 x 160 and stored as **static/must_logo.png**. The design of the logo is not changed.
- It sits on a white circle with 2 px padding so it stays readable on navy. Sizes: 84 px on the login page, 44 px in the sidebar and in the old console pages.

5. Layout and style

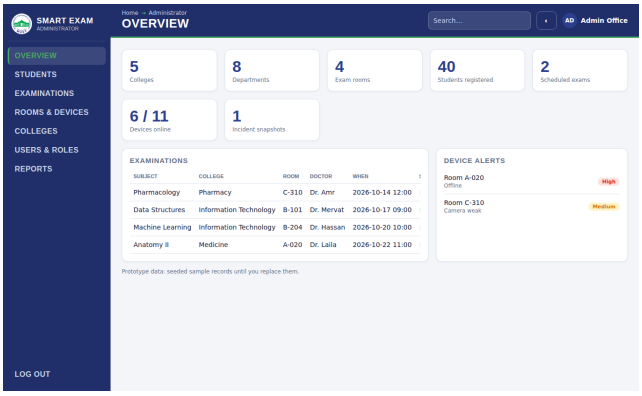
- **Login**: navy panel with logo, title "Smart Exam Monitoring System" (green highlight) and the human-review note; white sign-in card on the right.
- **Sidebar** 230 px, navy; active item green with a green bar; collapses to icons-only under 900 px.
- **Top bar** navy with a 3 px green bottom line, breadcrumb "Home -> Role", page title, search, theme toggle, user avatar.
- **Cards** white, 12 px corners, soft shadow. Buttons 8 px corners. Works on desktop, laptop and tablet.

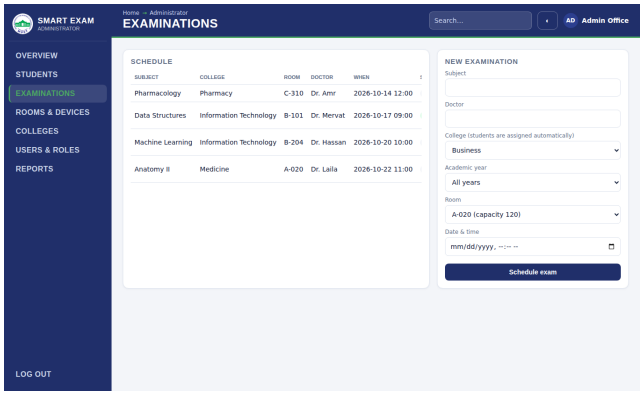


Login page



Doctor dashboard (live data from your Flask app)





Administrator overview

Administrator examinations

6. Screens and features

Doctor

Page	What it shows
Dashboard	Students detected, FPS, incidents to review, AI status, robot activity; live camera; student list with status; latest incidents
Live Monitoring	Live camera, Pause AI / Resume buttons (call /api/pause and /api/resume), student list, pause banner with the server alert message
Alerts & Incidents	Evidence snapshots from the evidence/ folder with time, event, link to the image, status; Review button with confirmation (saved in the database)
Robot Status	Papers detected, pan, tilt, sweeping state, robot camera feed, device status
My Students	Detected students as a seat grid, colored by normal / watch / suspected

Administrator

Page	What it shows
Overview	Colleges, departments, rooms, students, scheduled exams, devices online, incident snapshots; exam table; device alerts for offline or weak rooms
Students	Server-side search (top box), college filter, pagination (10 per page), Add student
Examinations	Schedule table with student counts and Cancel (with confirmation); form to create an exam: subject, doctor, college, year, room, date/time. Students of the chosen college/year are assigned automatically; a warning appears if they exceed room capacity
Rooms & Devices	Room, building, college, capacity, cameras, robot, status
Colleges	College cards with department, room and student counts; Add college; Add department
Users & Roles	Accounts and their access level (set in .env)
Reports	Incident snapshots for the last 7 days, exam status counts, Print/PDF, CSV export of exams and incidents, audit log of recent actions

Extra features added

- Audit log (who added, scheduled, cancelled or reviewed what)
- Confirmation dialogs for cancel and review
- Room-capacity warning when scheduling
- CSV export with protection against spreadsheet formula injection
- Dark / light theme, empty states and "no feed" state

7. Roles and security

- Session login at /login with two roles. Accounts and passwords come from the .env file (see .env.example). Old PROCTOR_USERNAME / PROCTOR_PASSWORD still work as the doctor account.
- Admin API routes return 403 for doctors; all other routes return 401 or redirect to /login without a session. Camera feeds, evidence images, /api/assign_student and WebSocket connections now also require login.
- Passwords are compared in constant time; SQL uses parameters only (tested with an injection string); the login redirect only accepts local paths.
- Change the default passwords and SECRET_KEY before any demo on a shared network, and use HTTPS if it leaves your laptop.

8. Database (SQLite, file smart_exam.db)

Table	Columns
colleges	id, name
departments	id, college, name
students	id, student_id (unique), name, college, department, year
rooms	id, code, building, college, capacity, cameras, robot, status
exams	id, subject, college, room, doctor, starts_at, status (Scheduled / Active / Completed / Cancelled)
exam_students	exam_id, student_id
alert_reviews	file, reviewed_by, reviewed_at
audit	id, at, user, action

Seed data (5 colleges, 8 departments, 4 rooms, 40 students, 4 exams) is sample data, created on first start. Delete smart_exam.db to reset. SQLite is fine for the prototype; the same tables move to PostgreSQL later.

9. API endpoints

Endpoint	Who	Purpose
GET /api/me	any	Name and role of the signed-in user
GET /api/status	any	Live tracks, FPS, robot, exam control (your existing route)
POST /api/pause, /api/resume	any	Pause or resume AI monitoring (existing)
GET /api/alerts	any	Evidence snapshots with reviewed flag
POST /api/alerts/{file}/review	any	Mark an incident reviewed
GET /api/exams	any	Exam list with student counts
GET /api/admin/overview	admin	Totals for the overview page
GET, POST /api/admin/students	admin	Search, filter, paginate; add student
POST /api/admin/exams, /{id}/cancel	admin	Schedule or cancel an exam
GET /api/admin/rooms	admin	Rooms and devices
GET, POST /api/admin/colleges; POST /departments	admin	Colleges and departments
GET /api/admin/users	admin	Accounts (no passwords)
GET /api/admin/reports	admin	7-day incidents, exam status, audit log
GET /api/admin/export/exams.csv, incidents.csv	admin	CSV downloads

10. Files delivered and how to install

File	Location	Change
dashboard.py	project root	Replaced: login, roles, new routes, database registration
admin_db.py	project root	New: database and REST API
templates/must_dashboard.html	templates/	New: final dashboard
templates/login.html	templates/	New: login page
templates/index.html	templates/	Re-themed operator console (now at /console)
templates/monitor.html	templates/	Re-themed; avatar CSS bug fixed (height:40 -> 40px)

File	Location	Change
static/must_logo.png	static/ (new folder)	MUST seal
.env.example	project root	Copy to .env and set real passwords

Run: `cd ~/Downloads/smart-exam && source venv/bin/activate && python dashboard.py`, then open http://127.0.0.1:5001 and sign in. No new pip packages are needed.

11. Testing done and known limits

- Tested here: Python syntax of both files; login, role checks (401/403), exam and student creation, duplicate IDs, cancel twice, SQL injection string, CSV export; JavaScript syntax; real browser rendering of login, doctor and admin pages with no script errors.
- **Not tested here:** your camera, YOLO and robot code (they need your Mac and hardware). The live feed shows "NO FEED" until dashboard.py is running with your camera.
- Robot battery is not shown because dashboard.py has no battery data. Add it to the robot state and one KPI card when the hardware reports it.
- Add-student and add-college use simple pop-up prompts; a proper form and a student details page are the next improvements.
- Barlow Condensed is a close match, not the confirmed MUST font. Arabic / right-to-left version not built yet.